Yixing Zhao

Curriculum Vitae

Department of Statistics and Actuarial Science
University of Waterloo
Waterloo, Ontario, Canada

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1 Education

Ph.D. in Statistics (Biostatistics) Sep 2024 - Present
University of Waterloo, Waterloo, Ontario, Canada
Supervisor: Prof. Liqun Diao. Expected completion: 2028.

Master of Mathematics in Statistics Sep 2023 - Aug 2024
University of Waterloo, Waterloo, Ontario, Canada

Bachelor of Mathematics in Computational Mathematics Sep 2020 - Jun 2023
University of Waterloo, Waterloo, Ontario, Canada
Minors: Statistics; Combinatorics and Optimization; Computing.

2 Research

Research interests: Bayesian methods and nonparametric inference; survival analysis and censored data; machine learning; mortality forecasting.

MAP-EM Polya Trees for Survival Analysis with Complex Censoring
Developed a unified Bayesian nonparametric method using a Polya tree prior for right-censored, current-status, and interval-censored event-time data. Derived a closed-form expectation-maximization algorithm for maximum a posteriori estimation and evaluated survival estimation through simulation studies.

Neural Regression Trees for Censored and Uncensored Outcomes
Research on conditional distribution estimation using a fixed binary-tree structure with neural-network-driven split locations and branch probabilities. The construction yields conditional density, distribution, and survival functions for exact and censored continuous outcomes.

3 Applied Research Experience

Research Intern May 2025 - Present
Mitacs Accelerate & Munich Re

- Developed post-COVID-19 mortality forecasts using historical cause-specific data, DSA variable selection, BMBE modeling, and age-segment selection; compared predictive performance with the Lee-Carter model.
- Implemented an R package for reproducible data processing, model fitting, validation, and prediction.
- Built a Bayesian MCMC model combining historical mortality shock patterns from World War II, HIV/AIDS, and the Spanish flu for pandemic scenario forecasting.

Research Assistant - Mortality Forecasting Winter 2024

4 Presentations

Zhao, Y., and Diao, L. "Neural Regression Trees for Censored and Uncensored Outcomes." Waterloo Student Conference in Statistics, Actuarial Science and Finance, University of Waterloo. *October 23, 2026 (upcoming).*

Zhao, Y., and Diao, L. "MAP-EM Polya Trees for Survival Analysis with Complex Censoring." 2026 SSC Annual Meeting, Hamilton, Ontario, Canada. June 16, 2026.

5 Teaching Assistantships

Graduate Teaching Assistant, University of Waterloo Sep 2023 - Present
Department of Statistics and Actuarial Science
Responsibilities include tutorials, student support, and assessment.

Course	**Terms**
STAT 431: Generalized Linear Models and their Applications	Winter 2026; Fall 2026
STAT 331: Applied Linear Models	Spring 2026
STAT 231: Statistics	Winter 2026; Spring 2026
STAT 221: Statistics (Non-Specialist Level)	Fall 2025
STAT 332: Sampling and Experimental Design	Fall 2024; Spring 2025
STAT 230: Probability	Spring 2025
STAT 437: Statistical Methods for Life History Analysis	Winter 2025
ME 202: Statistics for Engineers	Winter 2025
STAT 333: Stochastic Processes 1	Fall 2024

6 Awards

Women in Mathematics Mentorship Award	2025, 2026
University of Waterloo Graduate Scholarship	2025, 2026
Statistics & Actuarial Science Chair's Award	2024, 2026
International Master's Award of Excellence	2024

7 Academic Service and Mentoring

Mentor, Women in Mathematics Directed Reading Program Winter 2025; Spring 2026
University of Waterloo
Guided students in reading research papers and textbooks on Bayesian analysis and discussing applications in medicine and finance.

8 Technical Skills

Programming and tools: R, Python, SAS, C++, Git/GitHub.

Statistical computing: R package development, simulation studies, model validation, reproducible analysis.

Methods: Bayesian modeling, MCMC, nonparametric statistics, survival analysis, censored-data methods, variable selection, experimental design.